

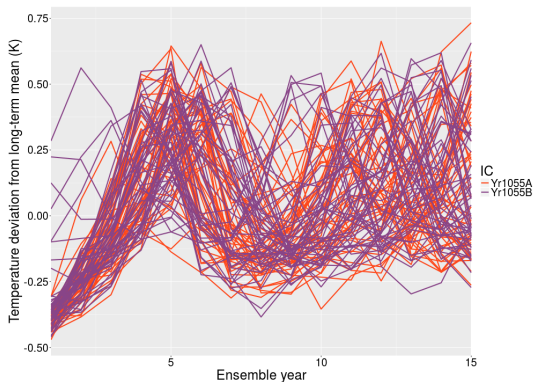
Climate Data Landscape

Brief introduction

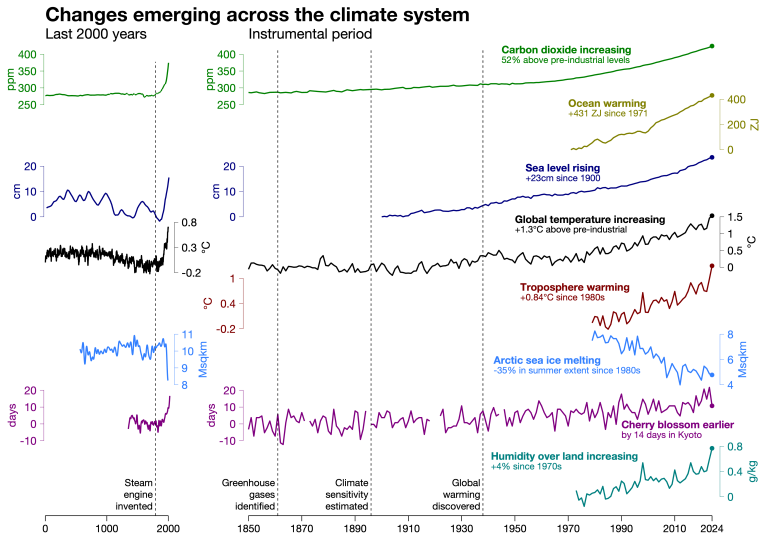
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EGS Building Rm 4.14

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Sources of climate data



Sources of climate data

3 main types (but not that simple)

① Observations

② Reanalysis

③ Models

Observations **Measurements** we take of the atmosphere and other components of the climate system

- From weather stations, rain gauges, satellites, ships, planes & other tools.
- Some are **raw**, others are collected into **gridded** (e.g. Worldclim) or interpolated products for ease of use.
- Gridded products use **statistical models**, cleaning and infilling for **spatially and temporally continuous data**
- As with all environmental data, come with **limitations**

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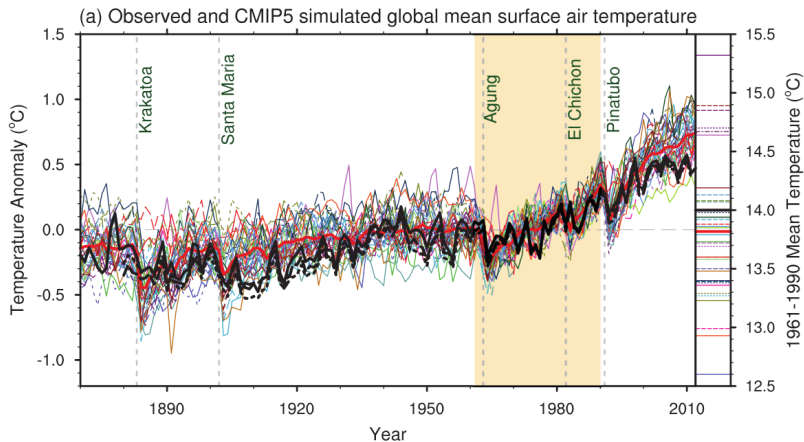
Reanalysis Latest weather prediction models rerun for short times with all available observations as input

- ERA5 state-of-the-art, has $\sim 25\text{km}$ grid (Table Mountain invisible)
- ERA5-Land only on land, but has $\sim 10\text{km}$ (Table Mountain a pyramid)

Model ensembles Such as CMIPs, CORDEX (RCMs)

Global temperature models vs observations

Essential that models are compared to observations and assessed for fitness-for-purpose (with cognisance of observation product limitations...)



Some valuable links:

Climate Change Model Projections IPCC (Intergovernmental Panel on Climate Change) Working Group 1 (WG1) 6th Assessment Report (AR6) [Interactive Atlas](#)

Weather station observations From SYNOP reports provided by meteorological services (such as the South African Weather Service (SAWS)) to World Meteorological Organisation (WMO) for initialisation of NWP:

- [Meteomanz.com](#); & [OGIMET.com](#)

General climate data portal + visualisations A few examples:

- [KNMI \(Dutch met service\) Climate Explorer](#)
- [US NOAA PSL plotting + analysis tool](#)
- [Zoom Earth](#) (includes satellite images)

Weather visualisations Windy.com only for current systems
[Earth Nullschool](#) provides similar animations of wind, with archived data from GFS NWP.

Re-Analysis

- General assessment: [Climate Reanalyzer](#)
- Temperature [Copernicus Climate Pulse](#)